A Quasi-Newton-like regularized approach to the order-value optimization problem[*]

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Abstract

The order-value function of order p returns, at each point, the pth smallest value among a finite family of functions $f_1, \dots, f_m$; minimizing it recovers the minimization of the minimum and of the maximum of the family in the extreme cases $p = 1$ and $p = m$, and provides a flexible tool for robust estimation, in which a prescribed number of outlying observations are discarded. We propose a regularized method for minimizing the order-value function subject to box constraints. At each iteration the method minimizes a strongly convex model that combines a piecewise-linear term built from the active gradients, a quasi-Newton curvature term defined by a symmetric positive semidefinite matrix, and a quadratic regularization; the curvature term carries second-order information without requiring explicit second derivatives, and a recently proposed first-order regularized method is recovered as the special case in which this term vanishes. We introduce an approximate optimality condition for the problem and prove that the method is well defined and attains the same worst-case iteration- and evaluation-complexity bounds as its first-order counterpart, so that the incorporation of curvature preserves the theoretical guarantees while offering the potential for higher-quality steps in practice. The applicability of the method to parameter-estimation problems with outliers is illustrated.

Keywords: Order-value optimization, regularized models, quasi-Newton methods, complexity, algorithms, applications.

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1 Introduction

{sec:intro}

Many optimization problems involve objective functions that are obtained not by aggregating a collection of simpler functions, but by selecting one of them according to its rank. The prototypical example is the order-value function of order p, introduced in [3, 4]: given m functions $f_1, \ldots, f_m$, its value at a point x is the pth smallest among $f_1(x), \ldots, f_m(x)$. Functions of this kind belong to the broader class of generalized order-value functions—those whose value at x is governed by order relations among the elements of a set $\{f_i(x)\}_{i \in I}$—a class systematized in [17]. Since the index that attains the pth position may switch abruptly as x moves, the order-value function is generally nonsmooth even when each f_i is continuously differentiable.

The extreme choices $p = 1$ and $p = m$ recover the minimum and the maximum of $\{f_i(x)\}$, respectively, so the order-value function interpolates between these two classical aggregates; the intermediate cases are the ones of practical interest. When x encodes portfolio positions and $f_i(x)$ is the loss incurred under scenario i, the order-value function coincides with the discrete value-at-risk (VaR) measure, a standard risk measure in financial risk management [16]. When $f_i(x)$ measures the misfit of a parametric model to the ith observation, minimizing the pth order-value function fits the parameters while discarding the $m - p$ worst-fitted observations. This makes order-value optimization a natural tool for robust estimation, where one seeks to remain insensitive to a minority of grossly corrupted data, or outliers.

The order-value optimization (OVO) problem was first formulated in [3], where a steepest-descent-type method was proposed and shown to converge to points satisfying a weak optimality condition. Sharper optimality conditions and a reformulation as a nonlinear program with equilibrium constraints were subsequently developed in [4]. A quasi-Newton method generalizing the scheme of [3] was introduced in [5], and a global optimization strategy combining multistart with a tunneling technique was proposed in [7]. On the computational-complexity side, [20] established that minimizing the order-value function over a polytope is NP-hard in the strong sense.

The systematic study of worst-case complexity in continuous optimization originates in the cubic-regularization analysis of [19], and complexity guarantees have since been obtained for a broad range of problems (see, e.g., [9, 10] and the monograph [13]). Within this line, [21] recently proposed a first-order method for box-constrained OVO that minimizes, at each iteration, a quadratically regularized piecewise-linear model of f, and established iteration- and evaluation-complexity bounds for reaching an approximate optimality condition.

In the present work we generalize the approach of [21] by enriching the regularized model with second-order information. At each iteration our method minimizes a model built from the same piecewise-linear convex term, a curvature term governed by a symmetric positive semidefinite matrix B, and a Tikhonov-type quadratic regularization. The matrix B is updated in a quasi-Newton fashion and therefore requires no explicit second derivatives, while the regularization renders the model strongly convex, so that each subproblem has a unique minimizer. Setting $B = 0$ recovers the first-order method of [21] exactly, so the proposed method is a strict generalization of it. We derive an approximate optimality condition for the box-constrained problem and prove that the method is well defined and attains the *same* worst-case iteration- and evaluation-complexity bounds as the first-order method. Thus the incorporation of curvature does not degrade the theoretical guarantees, while it offers the potential for higher-quality steps, and hence faster convergence, in practice—a behavior we illustrate on parameter-estimation problems in the presence of outliers.

The remainder of the paper is organized as follows. Section 2 defines the box-constrained order-value optimization problem and the proposed regularized method. Section 3 introduces an appropriate optimality condition and establishes the well-definiteness, convergence, and complexity of the method. Section **??** reports illustrative numerical experiments on parameter estimation in the presence of outliers. Final remarks are given in Section **??**.

Notation. The symbol $\|\cdot\|$ denotes the Euclidean norm. For $i = 1, \ldots, n$, $e_i \in \mathbb{R}^n$ denotes the ith column of the

2 A quasi-Newton regularized method

{sec:method}

Let $f_i : \mathbb{R}^n \to \mathbb{R}$, $i = 1, \ldots, m$, be given and let $p \in \{1, \ldots, m\}$ be fixed. The *order-value function of order* p is the mapping $f : \mathbb{R}^n \to \mathbb{R}$ defined by

$$f(x) = f_{i_p(x)}(x), \tag{1}$$ {ovo-function}

where $\{i_1(x), \ldots, i_m(x)\}$ is a permutation of $\{1, \ldots, m\}$ such that

$$f_{i_1(x)}(x) \le f_{i_2(x)}(x) \le \cdots \le f_{i_m(x)}(x); \tag{2}$$ {fi-sort}

that is, $f(x)$ is the pth smallest value among $f_1(x), \ldots, f_m(x)$. The boundary cases $p = 1$ and $p = m$ recover, respectively, the minimum and the maximum of $\{f_1(x), \ldots, f_m(x)\}$.

If $f_1, \ldots, f_m$ are continuous, then so is f. Differentiability, however, is not inherited: f may be nonsmooth at a point x at which the pth smallest value is attained by more than one index, i.e. whenever there exist $i \ne j$ with $f_i(x) = f_j(x) = f(x)$, since the active index $i_p(x)$ may then change as x varies.

We address the box-constrained order-value optimization problem

$$\underset{x \in \Omega}{\text{Minimize}}\, f(x), \tag{3}$$ {ovo-problem}

where $\Omega := \{x \in \mathbb{R}^n \mid \ell \le x \le u\}$ with $\ell, u \in \mathbb{R}^n$ and $\ell_i < u_i$ for $i = 1, \ldots, n$. Box constraints arise naturally in parameter estimation, where the unknowns are confined to physically or statistically meaningful bounds.

For $\delta > 0$ and $x \in \Omega$, define the *active index set*

$$I_\delta(x) := \{i \in \{1, \ldots, m\} \mid f(x) - \delta \le f_i(x) \le f(x) + \delta\}, \tag{4}$$ {idelta}

the set of indices whose values lie within a band of half-width δ around $f(x)$. As $\delta \to 0$, $I_\delta(x)$ shrinks to the set of indices that attain the order-value,

$$I_0(x) := \{i \in \{1, \ldots, m\} \mid f_i(x) = f(x)\}.$$

The method builds, around a reference point $\bar{x} \in \Omega$, a local model of f that combines the first-order information of the active components with curvature carried by a symmetric positive semidefinite matrix $B \in \mathbb{R}^{n\times n}$. In a quasi-Newton framework, B is updated from successive gradients, so that no second derivatives are computed explicitly. For given $\bar{x} \in \Omega$, $B \succeq 0$, $\delta > 0$, and $\sigma > 0$, the regularized model is

$$\Psi(x;\bar{x},B,\delta,\sigma) = \max_{i\in I_\delta(\bar{x})} \left\{ \nabla f_i(\bar{x})^T(x-\bar{x}) + \frac{1}{2}\left[(x-\bar{x})^T B(x-\bar{x}) + \sigma\|x-\bar{x}\|^2\right] \right\}. \tag{5}$$ {model}

The first term is the piecewise-linear convex function that captures the worst-case first-order behavior of f over the active components; the bracket adds the quasi-Newton curvature $(x-\bar{x})^T B(x-\bar{x})$ and a Tikhonov regularization $\sigma\|x-\bar{x}\|^2$. Since $\sigma > 0$, the model is strongly convex even when B is only positive semidefinite, so the subproblem (6) has a unique minimizer. For $B = 0$, the model reduces to the first-order regularized model of [21], which the present method thus generalizes.

Algorithm 2.1. Let $\delta > 0$, $\sigma_{\min} > 0$, $M > 0$, $\alpha \in (0,1)$, $\gamma > 1$, and $x^0 \in \Omega$ be given. Initialize $k \leftarrow 0$.

Step 1. Initialize $j \leftarrow 0$ and $\sigma_{k,j} = \sigma_{\min}$.

Step 2. Choose a symmetric positive semidefinite matrix $B_{k,j} \in \mathbb{R}^{n\times n}$ with $\|B_{k,j}\| \le M$ and compute $x^{k,j}_{\text{trial}}$ as the global minimizer of

$$\underset{x\in\Omega}{\text{Minimize}}\ \Psi(x;x^k,B_{k,j},\delta,\sigma_{k,j}). \tag{6}$$ {ovo-subpro}

Step 3. If

$$f(x^{k,j}_{\text{trial}}) \le f(x^k) - \alpha\|x^{k,j}_{\text{trial}} - x^k\|^2 \tag{7}$$ {ovo-armijo}

does not hold, set $\sigma_{k,j+1} = \gamma\sigma_{k,j}$, $j \leftarrow j+1$, and go to Step 2.

Step 4. Set $x^{k+1} = x^{k,j}_{\text{trial}}$, $\sigma_k = \sigma_{k,j}$, $j_k = j$, $k \leftarrow k+1$, and go to Step 1.

The bound $\|B_{k,j}\| \le M$ keeps the curvature from dominating the model. The freedom in choosing $B_{k,j}$ lets the method accommodate standard quasi-Newton updates such as BFGS or SR1, and the choice $B_{k,j} = 0$ for all k,j recovers the first-order method of [21].

The requirements on $x^{k,j}_{\text{trial}}$ are mild. The theory only needs $x^{k,j}_{\text{trial}}$ to be a stationary point of (6) with $\Psi(x^{k,j}_{\text{trial}};x^k,B_{k,j},\delta,\sigma_{k,j}) \le 0$; since $\Psi(x^k;x^k,B_{k,j},\delta,\sigma_{k,j}) = 0$, any descent method started at x^k produces such iterates. Moreover, (6) minimizes a strongly convex function over the compact convex set Ω, so it has a unique global minimizer and every stationary point is global; standard convex solvers apply directly.

3 Optimality condition and complexity

{sec:complexity}

Assumption A1. *The functions $f_1,\dots,f_m$ belong to $\mathcal{C}^1(\Omega)$; that is, they are continuously differentiable on Ω.*

We first show that a local minimizer of (3) also minimizes the model Ψ built at the corresponding reference point. These results connect the original nonsmooth problem with the convex subproblems solved at each iteration and underlie the optimality condition introduced below.

{teo:conv1}

Theorem 3.1. *Suppose that Assumption A1 holds. If x^* is a local minimizer of* (3)*, then it is also a local minimizer of Minimize*$_{x\in\Omega}$ $\Psi(x; x^*, 0, 0, 0)$.

Proof. Suppose not. Then there is $y^* \in \Omega$ with $\Psi(y^*; x^*, 0, 0, 0) < \Psi(x^*; x^*, 0, 0, 0) = 0$, i.e. $\nabla f_i(x^*)^T(y^* - x^*) < 0$ for all $i \in I_0(x^*)$. By Assumption A1, for each such i,

$$\lim_{t\to 0^+} \frac{f_i(x^* + t(y^* - x^*)) - f_i(x^*)}{t} = \nabla f_i(x^*)^T(y^* - x^*) < 0,$$

so there is $\bar{t}_i > 0$ with $f_i(x^* + t(y^* - x^*)) < f_i(x^*) = f(x^*)$ for $t \in (0, \bar{t}_i]$. With $\bar{t} = \min_{i\in I_0(x^*)} \bar{t}_i$,

$$f_i(x^* + t(y^* - x^*)) < f(x^*) \quad \text{for all } i \in I_0(x^*),\ t \in (0, \bar{t}]. \tag{8}$$ {bajaste}

For $j \notin I_0(x^*)$, continuity of $f_1, \ldots, f_m$ gives, for t small,

$$f_i(x^* + t(y^* - x^*)) < f_j(x^* + t(y^* - x^*)) \qquad \text{if } i \in I_0(x^*),\ f(x^*) < f_j(x^*), \tag{9}$$ {arriba}

$$f_i(x^* + t(y^* - x^*)) > f_j(x^* + t(y^* - x^*)) \qquad \text{if } i \in I_0(x^*),\ f(x^*) > f_j(x^*). \tag{10}$$ {abajo}

Hence the ranking positions of the active indices $I_0(x^*)$ within $\{f_i(x^* + t(y^* - x^*))\}_{i=1}^m$ are preserved for t small, so the pth position is still occupied by some $i \in I_0(x^*)$ and $f(x^* + t(y^* - x^*)) = f_i(x^* + t(y^* - x^*))$. With (8), $f(x^* + t(y^* - x^*)) < f(x^*)$ for t small. Since Ω is convex, $x^* + t(y^* - x^*) \in \Omega$, contradicting the local minimality of x^*. $\square$

{teo:conv2}

Theorem 3.2. *Suppose that Assumption A1 holds and let $B \succeq 0$ and $\sigma \geq 0$. If x^* is a local minimizer of* (3)*, then it is also a local minimizer of Minimize*$_{x\in\Omega}$ $\Psi(x; x^*, B, 0, \sigma)$.

Proof. Suppose not: there is $y^* \in \Omega$ with $\Psi(y^*; x^*, B, 0, \sigma) < 0$, i.e.

$$\max_{i\in I_0(x^*)} \{\nabla f_i(x^*)^T(y^* - x^*)\} + \tfrac{1}{2}\left[(y^* - x^*)^T B(y^* - x^*) + \sigma\|y^* - x^*\|^2\right] < 0.$$

Since $B \succeq 0$ and $\sigma \geq 0$, the bracket is nonnegative, so $\max_{i\in I_0(x^*)}\{\nabla f_i(x^*)^T(y^* - x^*)\} < 0$, that is, $\Psi(y^*; x^*, 0, 0, 0) < 0$, contradicting Theorem 3.1. $\square$

{teo:conv3}

Theorem 3.3. *Suppose that Assumption A1 holds and let $B \succeq 0$ and $\sigma, \delta \geq 0$. If x^* is a local minimizer of* (3)*, then it is also a local minimizer of Minimize*$_{x\in\Omega}$ $\Psi(x; x^*, B, \delta, \sigma)$.

Proof. Suppose not: there is $y^* \in \Omega$ with $\Psi(y^*; x^*, B, \delta, \sigma) < 0$. Since $I_0(x^*) \subseteq I_\delta(x^*)$, we have $\max_{i\in I_0(x^*)}\{\cdot\} \leq \max_{i\in I_\delta(x^*)}\{\cdot\}$, hence

$$\max_{i\in I_0(x^*)} \{\nabla f_i(x^*)^T(y^* - x^*)\} + \tfrac{1}{2}\left[(y^* - x^*)^T B(y^* - x^*) + \sigma\|y^* - x^*\|^2\right] \leq \Psi(y^*; x^*, B, \delta, \sigma) < 0,$$

that is, $\Psi(y^*; x^*, B, 0, \sigma) < 0$, contradicting Theorem 3.2. $\square$

Throughout, $\Sigma_q := \{\lambda \in \mathbb{R}^q \mid \sum_{j=1}^q \lambda_j = 1,\ \lambda_j \geq 0\}$ denotes the unit simplex. Following [21], the next consequence of Theorem 3.3 provides a first-order necessary condition for (3) and motivates the stopping criterion below.

{cor:exact}

Corollary 3.1. *Suppose that Assumption A1 holds and let $\delta \geq 0$. If x^* is a local minimizer of (3), then there exist $\mu \in \Sigma_{|I_\delta(x^*)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}^n_+$ such that $(\ell_i - x_i^*)\nu_i^\ell = (x_i^* - u_i)\nu_i^u = 0$ for $i = 1, \dots, n$ and*

$$\sum_{i \in I_\delta(x^*)} \mu_i \nabla f_i(x^*) + \sum_{i=1}^n (\nu_i^u - \nu_i^\ell) e_i = 0. \tag{11}$$

{exact-opt}

Proof. By Theorem 3.3 with $B = 0$ and $\sigma = 0$, x^* minimizes $x \mapsto \max_{i \in I_\delta(x^*)} \{\nabla f_i(x^*)^T (x - x^*)\}$ over Ω. Applying the KKT conditions for the minimization of a maximum of differentiable functions over a box [21, Lemma 3.1] yields multipliers $\mu \in \Sigma_{|I_\delta(x^*)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}^n_+$ with the stated complementarity and (11). $\square$

{def:optcond}

Definition 3.1. *Given $\epsilon > 0$ and $\delta > 0$, we say that $x \in \Omega$ satisfies the approximate optimality condition $C_{\delta,\epsilon}$ if there exist $\mu \in \Sigma_{|I_\delta(x)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}^n_+$ such that $(\ell_i - x_i)\nu_i^\ell = (x_i - u_i)\nu_i^u = 0$ for $i = 1, \dots, n$ and*

$$\left\| \sum_{i \in I_\delta(x)} \mu_i \nabla f_i(x) + \sum_{i=1}^n (\nu_i^u - \nu_i^\ell) e_i \right\| \leq \epsilon. \tag{12}$$

{miaprox-box}

Condition $C_{\delta,\epsilon}$ requires the gradient combination of the active components, corrected by the box multipliers, to be small in norm. For $\epsilon = 0$ it reduces to the exact condition (11) of Corollary 3.1. It serves as the stopping criterion for Algorithm 2.1.

{cor:step}

Corollary 3.2. *Suppose that Assumption A1 holds. Then, for every k and $j = 0, \dots, j_k$, there exist $\mu \in \Sigma_{|I_\delta(x^k)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}^n_+$ such that $(\ell_i - [x_{\text{trial}}^{k,j}]_i)\nu_i^\ell = ([x_{\text{trial}}^{k,j}]_i - u_i)\nu_i^u = 0$ for $i = 1, \dots, n$ and*

$$x_{\text{trial}}^{k,j} - x^k = \left(B_{k,j} + \sigma_{k,j} I\right)^{-1} \left[\sum_{i=1}^n (\nu_i^\ell - \nu_i^u) e_i - \sum_{i \in I_\delta(x^k)} \mu_i \nabla f_i(x^k) \right]. \tag{13}$$

{eqcoro1}

Proof. Subproblem (6) minimizes the strongly convex function $\Psi(\cdot\,; x^k, B_{k,j}, \delta, \sigma_{k,j})$ over the box Ω. By the KKT conditions for a maximum of differentiable functions over a box [21, Lemma 3.1], there exist $\mu \in \Sigma_{|I_\delta(x^k)|}$ and $\nu^\ell, \nu^u \in \mathbb{R}^n_+$ with the stated complementarity such that

$$\sum_{i \in I_\delta(x^k)} \mu_i \left[\nabla f_i(x^k) + (B_{k,j} + \sigma_{k,j} I)(x_{\text{trial}}^{k,j} - x^k) \right] + \sum_{i=1}^n (\nu_i^u - \nu_i^\ell) e_i = 0,$$

where $\nabla f_i(x^k) + (B_{k,j} + \sigma_{k,j} I)(x_{\text{trial}}^{k,j} - x^k)$ is the gradient at $x_{\text{trial}}^{k,j}$ of the ith component of Ψ. Since $\sum_i \mu_i = 1$, the curvature term factors out; as $\sigma_{k,j} > 0$ and $B_{k,j} \succeq 0$, the matrix $B_{k,j} + \sigma_{k,j} I$ is positive definite, and solving for $x_{\text{trial}}^{k,j} - x^k$ gives (13). $\square$

{a2}

Assumption A2. *For all k and $j = 0, \dots, j_k$, the multipliers ν^ℓ, ν^u of Corollary 3.2 satisfy $\max_{i=1,\dots,n} |\nu_i^\ell - \nu_i^u| \leq c_\nu$ for some constant $c_\nu > 0$ independent of k and j.*

The Mangasarian–Fromovitz constraint qualification guarantees that, for each (k, j), these multipliers are bounded [15]; Assumption A2 strengthens this to a single bound uniform in k and j, which is natural since Ω is compact and the active gradients are bounded under Assumption A1.

{a3}

Assumption A3. *There is a constant $c_\nabla > 0$ such that $\|\nabla f_i(x)\| \le c_\nabla$ for all $i = 1, \dots, m$ and all $x \in \Omega$.*

Assumption A3 holds automatically under Assumption A1, since Ω is bounded. Combining Corollary 3.2 with Assumptions A2 and A3 bounds the step. Since $B_{k,j} \succeq 0$, the eigenvalues of $B_{k,j} + \sigma_{k,j} I$ are at least $\sigma_{k,j}$, so $\|(B_{k,j} + \sigma_{k,j} I)^{-1}\| \le 1/\sigma_{k,j}$, and (13) gives

$$\|x_{\text{trial}}^{k,j} - x^k\| \le \frac{1}{\sigma_{k,j}} \left(\sum_{i=1}^{n} |\nu_i^\ell - \nu_i^u| + \sum_{i \in I_\delta(x^k)} \mu_i \|\nabla f_i(x^k)\| \right) \le \frac{n\, c_\nu + c_\nabla}{\sigma_{k,j}},$$

where the last inequality uses Assumptions A2, A3 and $\mu \in \Sigma_{|I_\delta(x^k)|}$. Thus

$$\|x_{\text{trial}}^{k,j} - x^k\| \le \frac{c_x}{\sigma_{k,j}}, \qquad c_x := n\, c_\nu + c_\nabla. \tag{14}$$ {ccotasub}

The step shrinks as $\sigma_{k,j}$ grows, which is the mechanism driving the acceptance of (7).

{a4}

Assumption A4. *There is $L > 0$ such that $\|\nabla f_i(y) - \nabla f_i(x)\| \le L\|y - x\|$ for all $i = 1, \dots, m$ and $x, y \in \Omega$.*

Assumption A4 is the standard gradient-Lipschitz condition. It implies, for all i and $x, y \in \Omega$,

$$f_i(y) \le f_i(x) + \nabla f_i(x)^T (y - x) + \tfrac{L}{2}\|y - x\|^2, \tag{15}$$ {taylor}

as well as the two-sided bound $|f_i(y) - f_i(x) - \nabla f_i(x)^T (y - x)| \le \frac{L}{2}\|y - x\|^2$. These make Ψ an upper model of f once $\sigma_{k,j}$ is large relative to L, which yields finite termination of the inner loop.

{lem1}

Lemma 3.1. *Let $a_1, \dots, a_r, b_1, \dots, b_r \in \mathbb{R}$ and $\beta > 0$ satisfy $a_1 \le \cdots \le a_r$, $b_j \le a_j - \beta$ for $j = 1, \dots, r$, and $b_{i_1} \le \cdots \le b_{i_r}$ for a permutation $\{i_1, \dots, i_r\}$ of $\{1, \dots, r\}$. Then $b_{i_q} \le a_q - \beta$ for $q = 1, \dots, r$.*

Proof. See [3, Lemma 2.1]. □

The next result gives a threshold on $\sigma_{k,j}$ beyond which the trial point is accepted at Step 3, so that the inner loop terminates after finitely many increases of $\sigma_{k,j}$, independently of k. The threshold scales with δ^{-1}: a narrower active band demands stronger regularization.

{teo:suff-desce

Theorem 3.4. *Suppose that Assumptions A1, A3, and A4 hold. Then, for all k and $j = 0, \dots, j_k$, if*

$$\sigma_{k,j} \ge \max\{L + 2\alpha,\ 9c_x^2/(2\delta)\}, \tag{16}$$ {este}

then $x_{\text{trial}}^{k,j}$ satisfies (7).

Proof. Assume (16); in particular $\sigma_{k,j} \geq L + 2\alpha$. By (15), for every $i \in \{1, \ldots, m\}$,

$$\begin{aligned}
f_i(x_{\text{trial}}^{k,j}) &\leq f_i(x^k) + \nabla f_i(x^k)^T(x_{\text{trial}}^{k,j} - x^k) + \tfrac{L}{2}\|x_{\text{trial}}^{k,j} - x^k\|^2 \\
&\leq f_i(x^k) + \nabla f_i(x^k)^T(x_{\text{trial}}^{k,j} - x^k) + \Big(\tfrac{\sigma_{k,j}}{2} - \alpha\Big)\|x_{\text{trial}}^{k,j} - x^k\|^2 \\
&= f_i(x^k) + \nabla f_i(x^k)^T(x_{\text{trial}}^{k,j} - x^k) + \tfrac{1}{2}\big[(x_{\text{trial}}^{k,j} - x^k)^T B_{k,j}(x_{\text{trial}}^{k,j} - x^k) + \sigma_{k,j}\|x_{\text{trial}}^{k,j} - x^k\|^2\big] \\
&\quad - \alpha\|x_{\text{trial}}^{k,j} - x^k\|^2 - \tfrac{1}{2}(x_{\text{trial}}^{k,j} - x^k)^T B_{k,j}(x_{\text{trial}}^{k,j} - x^k).
\end{aligned}$$

Since $B_{k,j} \succeq 0$, the last term is nonpositive and may be dropped. For $i \in I_\delta(x^k)$, the minimizer property $\Psi(x_{\text{trial}}^{k,j}; x^k, B_{k,j}, \delta, \sigma_{k,j}) \leq \Psi(x^k; x^k, B_{k,j}, \delta, \sigma_{k,j}) = 0$ gives

$$\nabla f_i(x^k)^T(x_{\text{trial}}^{k,j} - x^k) + \tfrac{1}{2}\big[(x_{\text{trial}}^{k,j} - x^k)^T B_{k,j}(x_{\text{trial}}^{k,j} - x^k) + \sigma_{k,j}\|x_{\text{trial}}^{k,j} - x^k\|^2\big] \leq 0,$$

so that

$$f_i(x_{\text{trial}}^{k,j}) \leq f_i(x^k) - \alpha\|x_{\text{trial}}^{k,j} - x^k\|^2, \qquad i \in I_\delta(x^k). \tag{17}$$ {cota-lem2}

On the other hand, by (14), the Cauchy–Schwarz inequality, and the two-sided consequence of Assumption A4, $|f_i(x_{\text{trial}}^{k,j}) - f_i(x^k)| \leq c_x^2/\sigma_{k,j} + Lc_x^2/(2\sigma_{k,j}^2)$ for all i; using $L \leq \sigma_{k,j}$, this is at most $3c_x^2/(2\sigma_{k,j})$, and (16) yields

$$|f_i(x_{\text{trial}}^{k,j}) - f_i(x^k)| \leq \tfrac{\delta}{3}, \qquad i = 1, \ldots, m. \tag{18}$$ {teo7eq1}

Consequently, any index with $f_i(x^k) < f(x^k) - \delta$ satisfies $f_i(x_{\text{trial}}^{k,j}) < f(x^k)$, and any index with $f_i(x^k) > f(x^k) + \delta$ satisfies $f_i(x_{\text{trial}}^{k,j}) > f(x^k)$. In particular $i_p(x_{\text{trial}}^{k,j}) \in I_\delta(x^k)$, and the number of components lying below the band is the same at x^k and $x_{\text{trial}}^{k,j}$. Write $I_\delta(x^k) = \{i_1, \ldots, i_r\}$ with $f_{i_1}(x^k) \leq \cdots \leq f_{i_r}(x^k)$, and let $\{i'_1, \ldots, i'_r\}$ reorder them so that $f_{i'_1}(x_{\text{trial}}^{k,j}) \leq \cdots \leq f_{i'_r}(x_{\text{trial}}^{k,j})$. Since the count below the band is preserved and $i_p(\cdot)$ lies in the band at both points, $i_p(x^k)$ and $i_p(x_{\text{trial}}^{k,j})$ occupy the same position q within the band; that is, $i_p(x^k) = i_q$ and $i_p(x_{\text{trial}}^{k,j}) = i'_q$. Applying Lemma 3.1 with $a_j = f_{i_j}(x^k)$, $b_j = f_{i_j}(x_{\text{trial}}^{k,j})$, and $\beta = \alpha\|x_{\text{trial}}^{k,j} - x^k\|^2$ —valid by (17)— gives $f_{i'_q}(x_{\text{trial}}^{k,j}) \leq f_{i_q}(x^k) - \beta$. Since $f_{i_q}(x^k) = f_{i_p(x^k)}(x^k) = f(x^k)$,

$$f(x_{\text{trial}}^{k,j}) = f_{i_p(x_{\text{trial}}^{k,j})}(x_{\text{trial}}^{k,j}) = f_{i'_q}(x_{\text{trial}}^{k,j}) \leq f(x^k) - \alpha\|x_{\text{trial}}^{k,j} - x^k\|^2,$$

which is (7). □

The inner loop thus terminates finitely at every outer iteration. The curvature affects only the constants below —through the bound M on $\|B_{k,j}\|$— and not the $O(\delta^{-2}\epsilon^{-2})$ iteration order, which coincides with that of the first-order method [21].

{teo:complexity

Theorem 3.5. *Suppose that Assumptions A1, A3, and A4 hold and that there is $f_{\text{low}} \in \mathbb{R}$ with $f(x) \geq f_{\text{low}}$ for all $x \in \Omega$. Then*

$$\sigma_k \leq \gamma \max\{L + 2\alpha,\ 9c_x^2/(2\delta)\}, \tag{19}$$ {cota-sigma}

and at most

$$\lfloor 1 + \log_\gamma(\sigma_k/\sigma_{\min}) \rfloor \tag{20}$$ {cota-fcnt}

function evaluations are performed per outer iteration to obtain (7)*. Moreover, the number of outer iterations k at which $C_{\delta,\epsilon}$ is not satisfied by x^{k+1} is at most*

$$\left\lfloor \frac{\big(M + \gamma \max\{L + 2\alpha,\ 9c_x^2/(2\delta)\}\big)^2}{\alpha} \cdot \frac{f(x^0) - f_{\text{low}}}{\epsilon^2} \right\rfloor. \tag{21}$$

{cota-iter}

Proof. Bounds (19) and (20) follow from Theorem 3.4 and the rule $\sigma_{k,j+1} = \gamma\sigma_{k,j}$: starting from $\sigma_{k,0} = \sigma_{\min}$, at most $\lfloor 1 + \log_\gamma(\sigma_k/\sigma_{\min})\rfloor$ increases occur before (16) holds, after which (7) is satisfied.

Let K be the set of outer indices k at which $C_{\delta,\epsilon}$ fails at x^{k+1}. Fix $k \in K$ and write $B_k = B_{k,j_k}$. By Corollary 3.2, $(B_k + \sigma_k I)(x^{k+1} - x^k) = \sum_{i=1}^n (\nu_i^\ell - \nu_i^u)e_i - \sum_{i \in I_\delta(x^k)} \mu_i \nabla f_i(x^k)$, hence

$$\big\|(B_k + \sigma_k I)(x^{k+1} - x^k)\big\| = \Big\| \sum_{i \in I_\delta(x^k)} \mu_i \nabla f_i(x^k) + \sum_{i=1}^n (\nu_i^u - \nu_i^\ell)e_i \Big\| > \epsilon,$$

the inequality holding because $C_{\delta,\epsilon}$ fails at x^{k+1}. Since $\|B_k\| \le M$,

$$\epsilon < \|(B_k + \sigma_k I)(x^{k+1} - x^k)\| \le (M + \sigma_k)\|x^{k+1} - x^k\|,$$

and with (19), $\|x^{k+1} - x^k\| > \epsilon/\big(M + \gamma \max\{L + 2\alpha, 9c_x^2/(2\delta)\}\big)$. As x^{k+1} satisfies (7),

$$f(x^{k+1}) \le f(x^k) - \alpha\|x^{k+1} - x^k\|^2 < f(x^k) - \frac{\alpha\epsilon^2}{\big(M + \gamma \max\{L + 2\alpha, 9c_x^2/(2\delta)\}\big)^2}.$$

Summing over $k \in K$ and using $f(x) \ge f_{\text{low}}$ gives $f(x^0) - f_{\text{low}} > |K| \cdot \alpha\epsilon^2/\big(M + \gamma \max\{L + 2\alpha, 9c_x^2/(2\delta)\}\big)^2$, from which (21) follows. □

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